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Training Impulse and Session Ratings of Perceived Exertion as Perceived by Athletes and Observed by Coaches During Preseason Football Camp

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Abstract

Smith-Hale, V, Mendoza, R, Vargo, C, Schoen, M, Jimenez, L, VanSumeren, M, Sabourin, J, Rider, BC, Martin, J, Moore, W, and Hew-Butler, T. Training impulse and session ratings of perceived exertion as perceived by athletes and observed by coaches during pre-season football camp. *J Strength Cond Res* XX(X): 000–000, 2025—Quantification of internal training load (TL), through wearable technology, represents an emerging metric for TL management in sport. The accuracy of subjective vs. objective assessments of TL, as perceived by athletes and observed by coaches, has yet to be evaluated during long-duration football practices (>2 hours). The purpose of this study was to evaluate relationships among objective heart rate (HR) and subjective ratings of perceived exertion (RPE) assessment of TL in collegiate American football athletes, and their coaches, during preseason camp. Twenty-four athletes (first-year, $n = 6$; returning, $n = 18$) and 2 coaches (football coach [FBC], $n = 1$; strength and conditioning coach [SCC], $n = 1$) participated. Athletes wore chest strap HR monitors during the first 6 days of preseason football camp for an objective TL assessment using training impulse (TRIMP: average exercise HR × duration of practice). Subjective TL was assessed by both athletes (perceived) and coaches (observed) using a session ratings of perceived exertion (sRPE; 0–10 rating scale; 10 = maximum effort). The average practice duration was 142.3 ± 35.5 minutes, with 139 total observations across 6 practices. Significant (alpha level $p \leq 0.05$) relationships were noted between TRIMP and sRPE from the athletes ($r = 0.62$), FBC ($r = 0.68$) and SCC ($r = 0.49$). When subdivided into first-year and returning groups, relationships were stronger between TRIMP and sRPE with first-year athletes (athlete $r = 0.68$; $p = 0.001$; FBC $r = 0.79$; $p = 0.001$; SCC $r = 0.64$; $p = 0.001$) than for returning athletes (athlete $r = 0.60$; $p = 0.001$; FB coach $r = 0.70$; $p = 0.001$; SCC $r = 0.40$: $p = 0.001$). These data suggest that subjective athlete perceptions and coach observations of TL are in low-moderate agreement, indicating these are not interchangeable, during long-duration preseason football camp.

Key Words: internal load, wearable technology, recovery

Introduction

American football is a high collision team sport characterized by short intermittent maximal-intensity bouts of exercise (20,40,41). Preseason training camp is a rigorous transition period, between off-season training to in-season competition, during which both training duration and intensity increase with minimal recovery time (20). The physical, mental, and metabolic demands of preseason training camp have shown to be greater than other training periods, including the 12-week competitive season (39). Not surprisingly, the high workloads sustained during preseason camp increase the rate of injury (36). Therefore, it is vital for both athletes and coaches to effectively monitor preseason camp workloads and make individual adjustments to both training loads (TL) and recovery rates to maximize performance gains and minimize the risk of injury.

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Precision training (i.e., the ability to specifically prescribe a TL during a practice session), largely driven through advances in wearable technology, has allowed athletes, coaches, and sport scientists to quantify TL and recovery among National Collegiate Athletic Association (NCAA) football athletes (23).The ability to monitor TL and recovery in football players is essential to determine players' response to training and better recognize underrecovery (e.g., elevated resting heart rate [RHR] and decreased heart rate [HR] variability) to decrease the likelihood of nonfunctional overreaching, injury, and illness (3).

Internal TL represents cumulative (acute and chronic) exercise intensity and can be measured objectively through HR monitors (19) or subjectively reported through ratings of perceived exertion (RPE) (2,15). The use of HR monitors has led to the use of Training Impulse (TRIMP), a method originally developed in 1991 by Banister (1) as a way to evaluate all physical components of training a single unit of physical effort. Training Impulse is calculated using the average HR measured during as a single training session (19,31) and produces valid and reliable scores of internal TL among individual (e.g., running and swimming) and

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team sport settings (e.g., soccer, basketball, and Canadian football) (7,15,21). Unfortunately, HR monitor systems can often be cost-prohibitive in sports with small budgets and large rosters (e.g., Division II or III NCAA football teams). Alternatively, a more cost-effective approach to measuring internal TL is through subjective self-reporting of session ratings of perceived exertion (sRPE), which was developed as a simple way for coaches to examine internal TL in a variety of training settings (e.g., weight training, practice, and game) (15). Session RPE is calculated by multiplying perceived intensity × duration (minutes) (15).

The ability of athletes and coaches to correctly perceive and observe internal TL during field training in large group settings (such as preseason football camp) has ramifications on athlete health and well-being. Any mismatch between coaches (who prescribe the TL) and athletes (who perform the TL) can result in undertraining, overtraining, illness and/or injury among players (4). Owing to the blunt collisions and varying environmental stressors (e.g., increased cardiovascular strain in warmer weather), characteristic of American football, it is vital that both football coaches (FBC; involved primarily with outdoor, game-related, activities) and strength and conditioning coaches (SCC; focused primarily on indoor, weight room, conditioning, activities) correctly interpret and integrate each athlete's TL to ensure that recovery is optimized for each individual player (23). In addition, potential differences between inexperienced first-year and experienced returning athletes may affect perceived (athlete) or observed (coaches) differences in athletes' RPE and recovery (23). For example, experienced players may hypothetically know what to expect from previous workouts and can "pace" themselves accordingly.

In addition to monitoring athletes' TL, it is of equal importance to understand how they recover from a training stimulus. To attain high-level performance over a long period of time, there must be an adequate balance between stress (training and competition load) and recovery (19,26,27). Such balance is also needed to decrease the likelihood of injury, illness, nonfunctional overreaching, and in some cases overtraining (18,26,27). During football preseason camp, it has been reported that athletes are at increased risk of injuries, particularly noncontact injuries in players with the highest workload ratios (36). Furthermore, such increases in injury rate have been associated with perceived increases in workloads (i.e., increased discomfort increases the subjective perception of workload intensity) that occur during the short time frame of football preseason training camp (10,11). Owing to the high workloads and limited time to recover during football preseason camp, it is important to monitor both HR-based measurements of TL and recovery responses both objectively and subjectively.

One increasingly popular objective measure of recovery is measurement of heart rate variability (HRV). Specific to football, HRV has been shown to be a validated, time-effective, and noninvasive way to monitor athletes' cardiovascular recovery between training bouts (13). The main advantage of using HRV to measure athlete recovery stems from the objectivity of measurement (i.e., quantification of the variance between heartbeats). Disadvantages include the cost of these monitors (especially for large teams, such as football) and the ability of athletes to truly achieve "resting" conditions within (chaotic) field settings. Therefore, to date, there has been limited research conducted simultaneously on HR-based measurements of both TL (HR) and recovery (HRV) with football athletes.

The primary aim of this study was to compare objective HR-based measurements with subjective sRPE assessments of internal TL in a cohort of collegiate American football athletes and 2 different coaches (FBC and SCC) during the first week of preseason training camp. A secondary aim was to examine the potential differences between first-year and returning athletes' internal TL. A tertiary aim of this study was to explore relationships between internal TL and recovery using the HR-based monitoring algorithms during preseason camp.

Methods

Experimental Approach to the Problem

This prospective study occurred during the first 6 official days of preseason summer football camp, with outcome measures and procedures summarized in Figure 1. During 6 consecutive preseason practices, sRPE was recorded to analyze athletes' perception of TL and coaches' observation of athletes' TL. To validate athletes perceived and coaches observed TL, athletes wore chest strap HR monitors during each practice session to capture internal TL. In addition, across all practices, recovery measures (HRV and RHR) were obtained and standardized for time (i.e., every morning and before practice) and location (i.e., on the floor of the weight room). Both players and coaches were blinded to the results captured by the HR monitors for the duration of the study. Internal load and recovery measures were compared between first-year and returning players to further examine the relationship and differences between the 2 groups' (first-year vs. returning) TL and recovery responses during consecutive preseason camp practices.

Subjects

A total of 29 NCAA Division II American football athletes originally participated in this study. Inclusion criteria included those athletes selected by the coaching staff to wear HR monitors during practice for health and safety purposes. Exclusion criteria included those athletes, assigned HR monitors, with incomplete data on 2 of the 6 practices (33%) for a variety of reasons including forgetting to wear the monitor, equipment malfunction, or injury. We chose this exclusion criteria (66% adherence rate) as it reflects previous adherence rates to exercise studies, associated with a wide range of subject attrition rates (25–50%) (29). As such, 5 of the 29 athletes missed 2 or more of the 6 practices and were excluded from the final analysis. Thus, 24 athletes' data were used in the final analysis; they were grouped as first year on the team ($n = 6$) or returning player ($n = 18$). Physical characteristics of the players are included in Table 1. Only 2 coaches, 1 FBC (>10 years' experience with this football program), and 1 SCC (<1-year experience with this football program) agreed to participate in this study. Athletes self-reported age (from birthdate) was 20 years ($M = 20$, $SD = 2$). The study was approved by an IRB at Wayne State University's Institutional Review Board for Human Subject's Research and adhered to the ethical standards of the Declaration of Helsinki. All athletes (18 years and older) provided written informed consent after details of the study and potential risks were communicated verbally and in written form before data collection.

Procedures

Anthropometric Assessments. On the same day, during the first week of preseason camp, body composition measurements were obtained by dual-energy x-ray absorptiometry. All scans were

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Figure 1. Study summary.

completed using a Hologic Horizon A, (Marlborough, MA) with TBAR1209 NHANES BCA calibration (completed each day) and analyzed using (APEX System Software version 5.6.0.5) to determine whole body levels of lean mass, fat mass, bone, and body fat percentages. All scans were performed and analyzed by the same operator to maintain consistency. Athletes wore compression shorts without any metal, such as buttons or zippers and removed any jewelry before testing. Athletes' height and weight was measured using a stadiometer and electronic scale (Seca 703 scale, Chino, CA). In addition, athletes were instructed to refrain from exercise and food for 4 hours before testing.

Recovery Assessments. Athletes reported to the weight room approximately 60–90 minutes (~7:30–8:00 AM) before the start of practice at 9:00 AM Each athlete secured a preassigned chest strap HR monitor (Firstbeat, Jyväskylä, Finland) around their chest and lay supine for 3 minutes to obtain HRV and RHR measures. Firstbeat software system (Firstbeat, version 2.8.1) reported HRV as the root mean square of the successive difference and RHR variables. Such short-duration (i.e., 1–5 minutes) supine measurements of HRV is the preferred method for field measurements (6). Data were automatically uploaded to a web-based platform for analysis. After HRV and RHR measurements were collected, athletes continued to wear these same chest strap HR monitors to measure TL until the end of practice.

Internal Training Load. During all practices, players' HR was captured by radiotelemetry (Firstbeat Team, Firstbeat HR plastic belt, Jyväskylä, Finland). The sampling rate for the chest strap monitor was 125 Hz (125 times per second) with an accuracy of RR-interval detection of 2 milliseconds (34).

Table 1

Physical characteristics of American football players.*

	First year (n = 6)	Returner (n = 18)	Combined (n = 24)
Age (y)	18.33 ± 0.48	21.0 ± 1.34	20.33 ± 1.65
Height (m)	1.88 ± 0.03	1.88 ± 0.04	1.87 ± 0.04
Body Mass (kg)	123.90 ± 7.69	113.90 ± 19.40	116.21 ± 17.67
BMI (kg·m^{-2})	34.87 ± 1.75	32.63 ± 5.73	33.16 ± 5.11
Body fat (%)	30.82 ± 2.07	25.6 ± 5.91	29.96 ± 5.67
Lean mass (kg)	82.42 ± 6.22	79.49 ± 8.61	80.25 ± 8.13
BMD (g·cm^{-2})	1.33 ± 0.12	1.4 ± 0.08	1.38 ± 0.09

*Values are mean ± *SD*.

The chest strap monitors were continuously monitored by the research team and adjusted only when a signal was temporarily lost (i.e., the chest strap needed to be tightened to remain at the chest level). The data were downloaded using the Firstbeat web-based platform to determine TL. Two internal TL methods were assessed with the first one being TRIMP, which was calculated by Firstbeat software version 2.8.1 using the HR reserve method. The second measure of internal TL was calculated using Edwards' TRIMP (15, 42). For validity purposes, Edwards' TRIMP was also calculated by measuring the product of the accumulated training duration (minutes) of 5 HR zones by a coefficient relative to each zone (50–59% of HR_{max} = 1, 60–69% HR_{max} = 2, 70–79% HR_{max} = 3, 80–89% HR_{max} = 4, 90–100% of HR_{max} = 5) and then adding the results (21,25). Heart $rate_{max}$ was estimated using the equation (220—Age). Edwards' TRIMP produces valid and reliable measurements of internal load for high intensity-based sports (7).

Session Ratings of Perceived Exertion. At the end of practice, each athlete reported their sRPE using the CR10-scale modified by Foster et al. (16). All athletes and coaches were familiarized with the scale before the study began, and they were asked to report which number best described the whole session and not just the end of practice. Both the familiarization and data collection were performed by a single (i.e., the lead) researcher. Owing to the athletes' rigorous camp schedule, sRPE was collected right after practice (5–10 minutes), as each athlete exited the football field (23). Both FBC and SCC were asked to report their observed sRPE, for each individual athlete, which was also collected using the same scale immediately after practice. To separate athletes' perceptions and each coach's observations, each coach was given a list of all the study subjects (in alphabetical order, by the last name) to report their observed sRPE for each individual player. Foster et al. determined TL by multiplying training duration (minutes) by reported sRPE. These same methods were used for the purpose of this study; where athletes' perceived sRPE × minutes of practice = $ATHLETE_{TL}$ and each coach observed sRPE × minutes of practice = Coach TL. Each coach will be denoted as SCC (SCC_{TL}) or FBC (FBC_{TL}).

Statistical Analyses

The relationships among various TL measurements $ATHLETE_{TL}$, SCC_{TL}, FBC_{TL}, TRIMP, Edwards' TRIMP, and recovery

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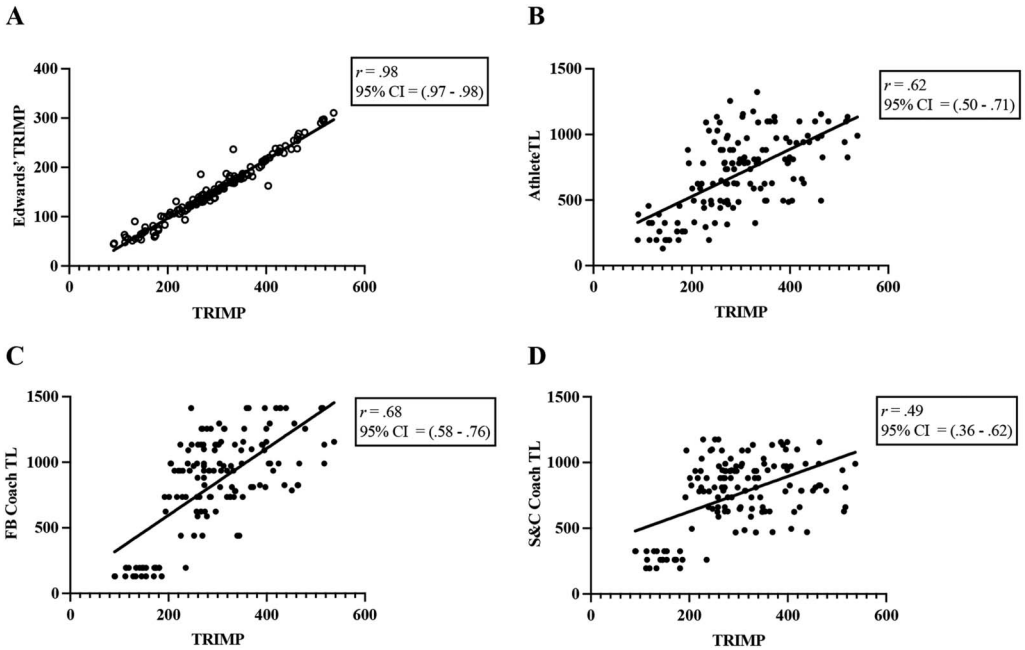

Figure 2. Correlations between heart rate internal training load and sRPE. Relationship between all observations ($N = 129$) for Edwards TRIMP and TRIMP (A), athlete TL and TRIMP (B), FB coach TL and TRIMP (C), and S&C coach TL and TRIMP (D). Correlation coefficient is represented by r ($p \leq 0.05$).

variables (HRV and RHR) were analyzed using Pearson's correlation coefficient (r), with the interpretation of variance using the coefficient of determination (R^2). The coefficient of determination is the correlation squared, and this indicates the amount of variance shared among 2 variables. More specifically, this will help assess whether coaches observation or athletes perception of practice load would give similar results from internal load monitoring using HR monitors. Cohen's d effect sizes (ES) were calculated to explore the magnitude in the mean between TL variables. Effect sizes were interpreted as follows: <0.2, trivial; 0.2–0.59, small; 0.6–1.19, moderate; 1.2–1.9, large; and >2.0, very large (8). All variables were found to be normally distributed, as assessed by Shapiro-Wilk's test ($p > 0.05$), except $ATHLETE_{TL}$ and both coaches (SCC and FBC) observed TL. Therefore, to examine the differences between first-year and returning athletes' TL and recovery responses across preseason camp, a nonparametric Mann-Whitney U-test was conducted. Percent of maximum score (POMS) was applied to HRV and TRIMP variables to facilitate visually examining all variables on the same scale to observe patterns in TL and recovery. It should be noted that the sample size is small and different group sizes are very common in applied field work. The results are expressed as mean (M) ± SD unless otherwise noted. Statistical significance was set using an alpha level of $p \leq 0.05$. IBM SPSS Statistics for Macintosh, version 26.0 (Armonk, NY) was used for statistical analyses.

Results

Heart rate-based TL and both perceived and observed TL accumulated 139 total observations across the 24 athletes' participation in 6 practices (142.26 ± 35.51 minutes). Correlations across all observations of TL ($ATHLETE_{TL}$, SCC_{TL}, FBC_{TL}, TRIMP, and Edwards' TRIMP) were all significant ($p \leq 0.01$) with the strongest relationships being observed between Edwards' TRIMP and all TL measures (Figure 1). Training Impulse and Edwards's TRIMP reported a strong correlation ($r = 0.98$, $p = 0.01$; ES = 1.72, large). Thus, for consistency and generalizability of internal TL, Edwards's TRIMP was chosen to reflect all HR-based internal TL as the 2 values were statistically interchangeable (i.e., $r = 0.98$; similar correlation coefficients and p-values) (Figure 2).

Across the other measures of TL, the strongest relationship was reported between Edwards' TRIMP and the FBC_{TL} ($r = 0.68$, $p = 0.01$; ES = 1.98, large), where the weakest relationship was between Edwards' TRIMP and the SCC_{TL} ($r = 0.49$, $p = 0.01$; ES = 2.26, very large) (Figure 2). Linear relationships were also found between internal TL and recovery variables. Small negative relationships between HRV and Edwards' TRIMP ($r = -0.29$, $p = 0.001$; ES = 3.06, very large), $ATHLETE_{TL}$ ($r = -0.20$, $p = 0.02$; ES = 3.01, very large), SCC_{TL} ($r = -0.20$, $p = 0.02$; ES = 3.63, very large), and FBC_{TL} ($r = -0.17$, $p = 0.04$; ES = 3.55, very large). Conversely, RHR had small positive relationships with Edwards' TRIMP ($r = 0.22$, $p = 0.01$; ES = 3.26, very large) and SCC_{TL} ($r = 0.29$, $p = 0.001$; ES = 3.63, very large).

Data were then subdivided by first-year and returning players. Stronger relationships were observed for all TL variables among first-year athletes and all compared with returning athletes (Figure 3). First-year players HRV were significantly negatively related to all TL variables: Edwards' TRIMP ($r = -0.58$, $p = 0.001$; ES = 2.98, very large), $ATHLETE_{TL}$ ($r = -0.49$, $p = 0.003$; ES = 3.02, very large), SCC_{TL} ($r = -0.40$, $p = 0.02$; ES = 4.11, very large), and FBC_{TL} ($r = -0.55$, $p = 0.01$; ES = 2.36, very large). Whereas RHR for first-year athletes was significantly, positively related to Edwards' TRIMP ($r = 0.49$, $p = 0.003$; ES = 3.04, very large), $ATHLETE_{TL}$ ($r = 0.42$, $p = 0.01$; ES = 3.04, very large), and FBC_{TL} ($r = 0.39$, $p = 0.02$; ES = 2.38, very large). However, returning athletes only displayed significant correlations between HRV and Edwards' TRIMP ($r = -0.20$, $p = 0.05$; ES = 3.16, very large) and RHR and SCC_{TL} ($r = -0.22$, $p = 0.03$; ES = 3.10, very large).

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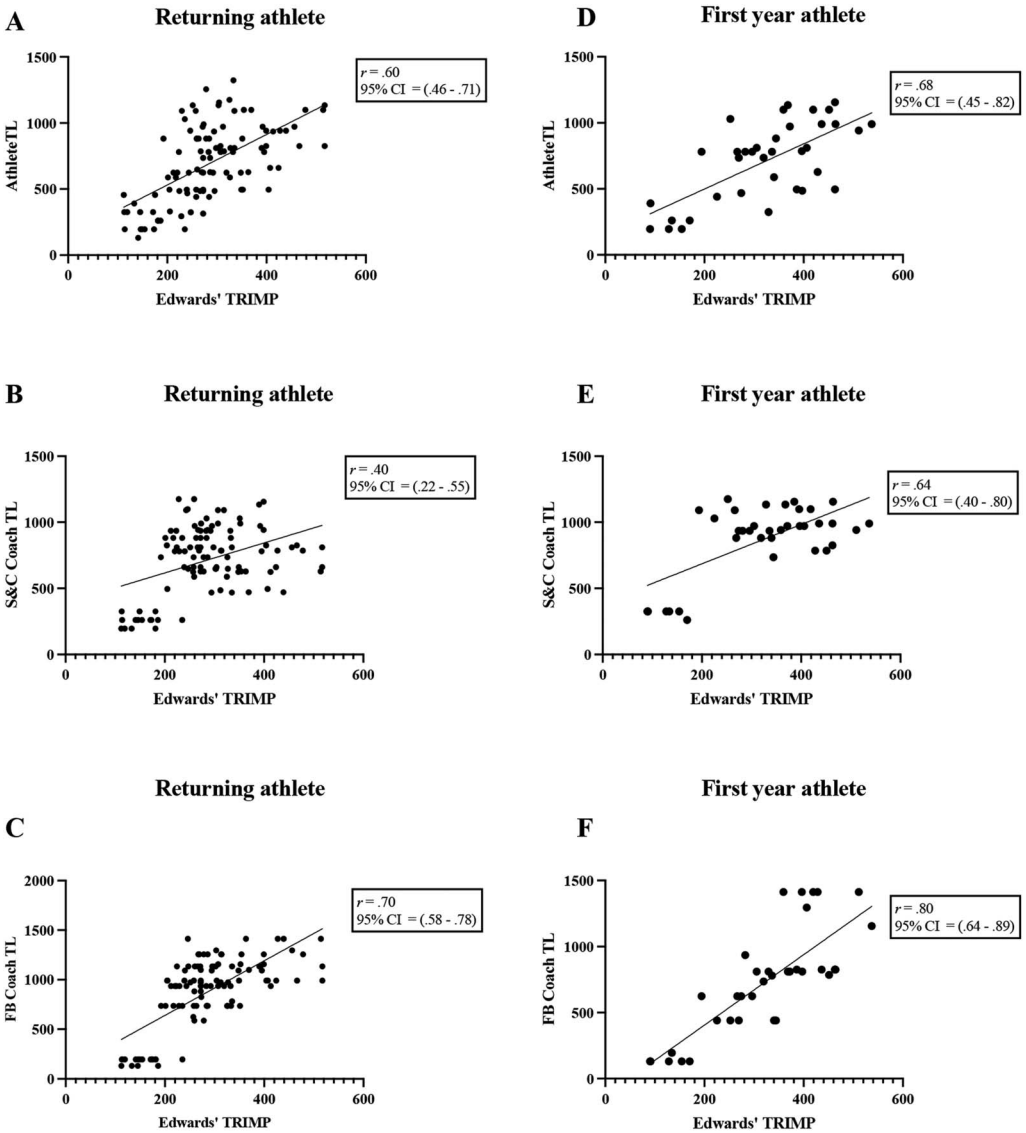

Figure 3. Correlations between first-year and returning athletes internal TL and sRPE TL. Relationship between returning (n = 6) and first-year (n = 18) NCAA football athletes internal training load Edwards TRIMP and self-report training load by athletes and coaches. Returning athletes Edwards TRIMP and athlete TL (A), S&C coach TL (B), and FB coach TL (C). First-year athlete Edwards TRIMP and athlete TL (D), S&C coach TL (E), and FB coach TL (F). Correlation coefficient is represented by r ($p \leq 0.05$).

Next, median differences by athlete experience was examined with the Mann-Whitney U test. Significant differences in HRV existed between first-year (Mdn = 74.56) and returning athletes (Mdn = 78.43) across preseason practices (U = 2,329, z = 2.36, p = 0.018, r = 0.199) (Figure 4A). First-year RHR (Mdn = 66 bpm) was also found to be significantly higher than returning athletes (Mdn = 60) (U = 1,106, z = −3.52, p = 0.001, r = −0.29) (Figure 4B). Edwards' TRIMP was significantly different between the 2 groups (U = 1,406, z −2.15, p = 0.03, r = −0.18) (Figure 5A); first-year (Mdn = 780) athletes accumulated a greater load compared with returning players (Mdn = 697). However, the athletes' perception of TL did not significantly differ between first-year (Mdn = 780) and returning athletes (Mdn = 735) (U = 1795, z = −0.284, p = 0.776, r = −0.02) (Figure 5B). Furthermore, both SCC_{TL} (U = 1,058, z = −3.77, p = 0.001, r = −0.32) and FBC_{TL} (U = 2,346, z = 2.37, p = 0.018, r = 0.20) observed differences between first-year and returning athletes TL (Figure 5C). On average, the SCC observed first-year athletes to have a greater TL (869.78 ± 274.08) than returning players (714.80 ± 257.93). Whereas the FBC observed first-year athletes to have a lower TL (735.36 ± 396.63) than returning players (874.89 ± 362.77).

Figure 6 allows for visual inspection of all combined main outcome variables, in both first-year athletes (6A) and returning athletes (6B), converted into a relative rating scale (0–10; POMS) for each day of data collection (i.e., time). These variables include objective internal TL and HRV assessments as well as subjective sRPE assessments for the athletes and coaches, all reported as mean values. This visual (nonstatistical) representation highlights the statistical differences between first-year vs. retuning athletes, previously documented in Figures 4 and 5, while demonstrating the inverse relationship between TL and recovery using the HR-monitoring system over time.

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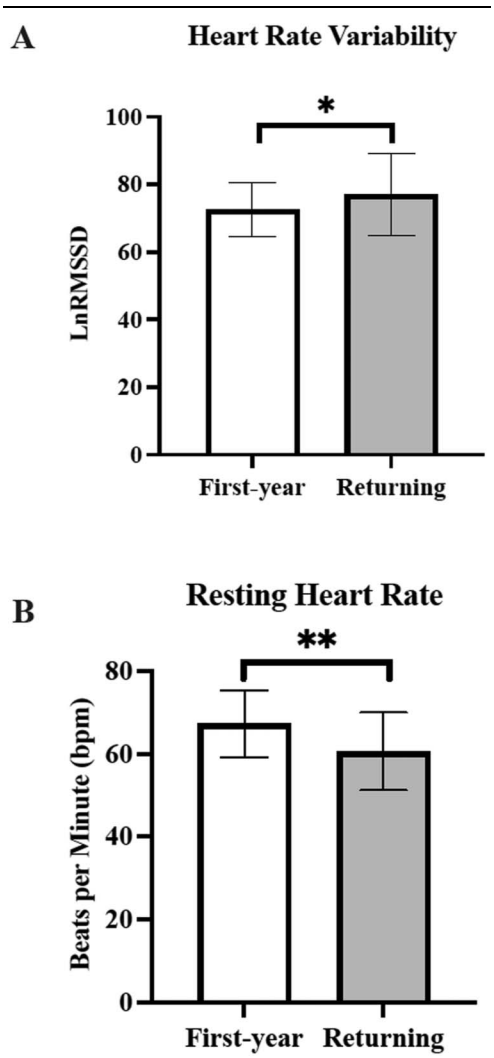

Figure 4. Differences between first-year and returning athletes recovery. LnRMSSD = natural log of root mean successive differences × 20. Asterisks indicates significant differences between first-year and returning player, $^{*}p < 0.05$; $^{**}p < 0.001$. First-year (n = 6); returning player (n = 18).

Discussion

The primary aims of this study were to examine the relationships among American football athletes' internal TL from 3 different measurements that included athletes' perception ($ATHLETE_{TL}$), coaches' observation (FBC_{TL} and SCC_{TL}), and HR-based (Edwards' TRIMP) TL. Athletes (R^2 = 0.39; ES = 1.84, large), as well as both the football (R^2 = 0.47; ES = 1.98, large) and S&C (R^2 = 0.17; ES = 2.26, very large) coaches, were all able to predict internal TL, which aligned with objective HR-based measures. Notably, the SCC's perception had the lowest correlation with athlete internal load (17% of variance), although the relationship remained statistically significant. Collectively, these significant relationships suggest that both subjective athlete perception, coach observation, and athlete internal load are in agreement during long-duration (>2 hours) preseason football camp settings. However, the low to moderate strength of these correlations suggests that a variety of other internal and external athlete stressors differentially affect HR and effort perception for both coaches and athletes, which requires more critical psychobiological scientific dissection.

Researchers have found sRPE methodology to be an inexpensive, efficient, and reliable way to measure internal TL in place of expensive and more invasive HR monitoring (7,15,21). There are moderately strong relationships between objective and subjective measurements of internal load in Canadian football (7), basketball (30), and soccer (21) players with correlation coefficients ranging between 0.68 and 0.85. The results from this study found a slightly lower correlation coefficient in our NCAA DII American football players (r = 0.62), but still in reasonable agreement with previous studies. A possible explanation for the lower correlation coefficient documented in this study is that the sRPE load ratings included other physiological stressors (i.e., ambient environment, and psychological stress) imposed on the athletes during preseason camp that might not have been captured in previous studies that were observed over an entire football season (7) and over multiple weeks of preseason camp (22). In addition, to further explain the differences in correlations between sports, there are specific factors to the sport of football that could influence players perception of load. First, football players train with large amounts of protective equipment and the added equipment could explain the difference between this study and other outdoor field sports such as soccer (9). In addition, it is possible that the intense collisions in football could influence player perception of the load experienced compared with other sports such as basketball (7,40).

In addition to assessing relationships between objective and subjective assessments of internal TL in athletes, in this study, we examined the main FBC and SCC's ability to quantify internal TL in athletes. Similar studies, seeking to investigate whether sport coaches can accurately determine athletes internal load, have been performed in soccer (4), volleyball (35), and basketball (28), but not with American FBCs. The results from this study highlight that the FBC was more accurate in observing the athletes' internal TL with a moderate correlation (r = 0.68). This moderately strong correlation aligns with previous studies (r = 0.70–0.84), which have examined coaches' observed TL and athletes' TL with swimming (38), runners (17), and volleyball (35).

To the best of our knowledge, no researchers have examined an SCC observation of TL during sports practice. This study documented a weaker, but still statistically significant, correlation for the SCC (r = 0.41) compared with the FBC's observation of athletes' internal TL. Brink et al. (4) found a similar weak relationship between soccer coaches' observed TL and soccer athletes' perceived TL (r = 0.41). Although the Brink et al. (4) study did not examine the SCC, it is interesting that similar relationships were found. Field-based practices are programmed and ran by the sport coaches (i.e., the FBC) and not the SCCs. Thus, it is possible that the SCC examines TL differently for field-based practices than they do for weight room-based programming (which they provide direct oversight). This exploratory finding highlights the need for SCCs to better understand the cumulative TL athletes undergo during field-based practices to help manage overall TL. Future researchers should continue to examine the SCCs ability to observe internal TL during both weight training and conditioning sessions, as well as sport training sessions; such research should help to better coordinate load management strategies involving teams of sport coaches, medicine, nutritionists, and scientists moving forward.

The secondary aim of this study was to evaluate potential differences between first-year and returning football athletes load throughout the first 6 days of preseason camp. There were positive correlations between Edwards' TRIMP and $ATHLETE_{TL}$, FBC_{TL}, and SCC_{TL}, and they were stronger in the first-year athletes than the returning athletes. These findings were unexpected,

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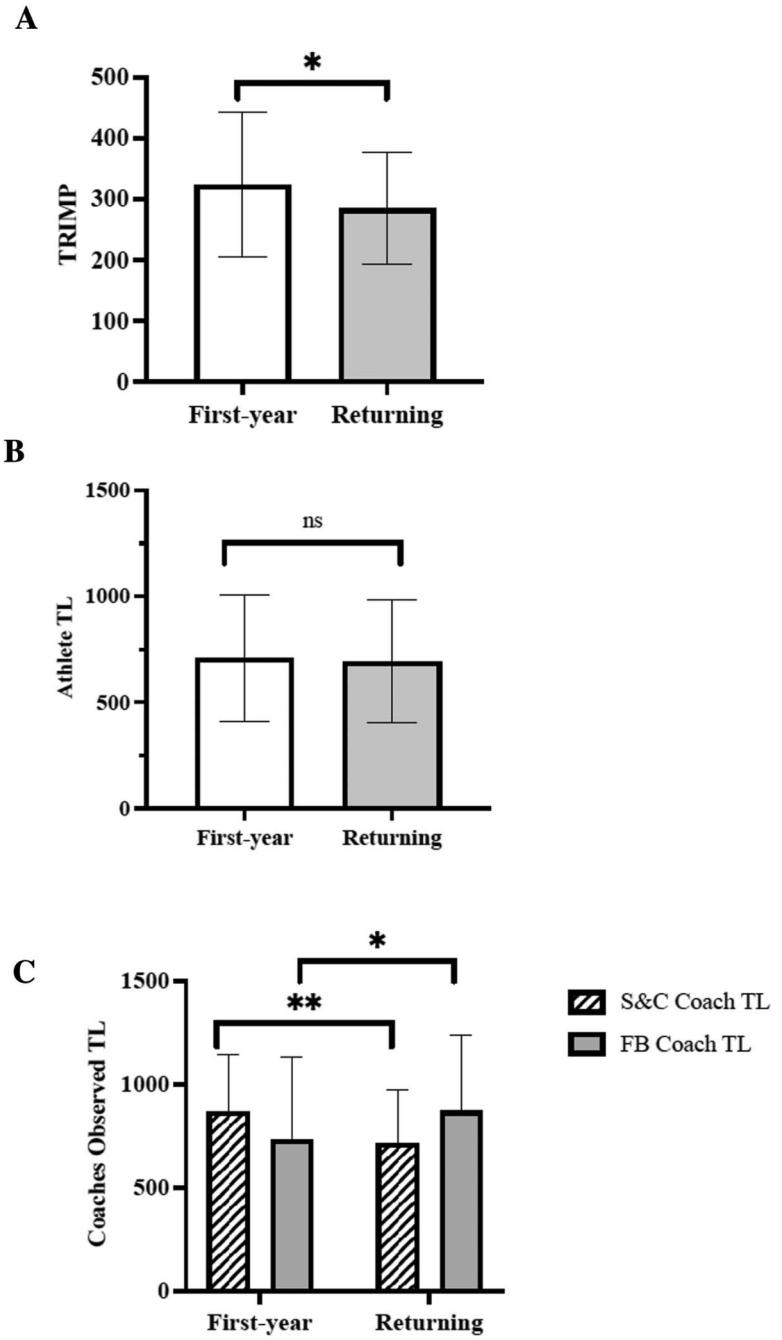

Figure 5. Differences between first-year and returning athletes training load. S&C coach TL = strength and conditioning coach training load, FB coach TL = football coach training load. Asterisks indicates significant differences between first-year and returning athletes, $^{*}p < 0.05$; $^{**}p < 0.001$; ns = $p > 0.05$ (not significant).

as the first-year players were unfamiliar with the training and personnel, so we expected weaker correlations between HR values (Edwards TRIMP) and subjective ratings of internal load between $ATHLETE_{TL}$ and $COACH_{TL}$ ratings. One possible reason why the perceived and observed relationships with HR objective metrics was weaker in returning players might be explained by "muscle memory." For example, perceived effort may be modified by memory-induced alterations, which could affect pacing strategies during subsequent long-duration workouts (32,33) or through augmented neural replay and wakeful consolidation of motor skills (5), which may confound ratings of perceived effort in the returning players, but not the first-year players. Alternatively, researchers have previously found that compared with returning athletes, first-year college football athletes have the lowest BW (128 ± 12.7 kg), greatest %BF (22.5 ± 12.7%), and perform the poorest in fitness tests such as max bench press, squat, and power clean (22). It is thereby possible that differences in anthropometrics and physical fitness could make it easier for coaches to make a more accurate assessment on first-year athletes' load. In fact, Wade et al. (37) highlighted that first-year college athletes lack both physical and mental toughness to endure the rigors of collegiate sport

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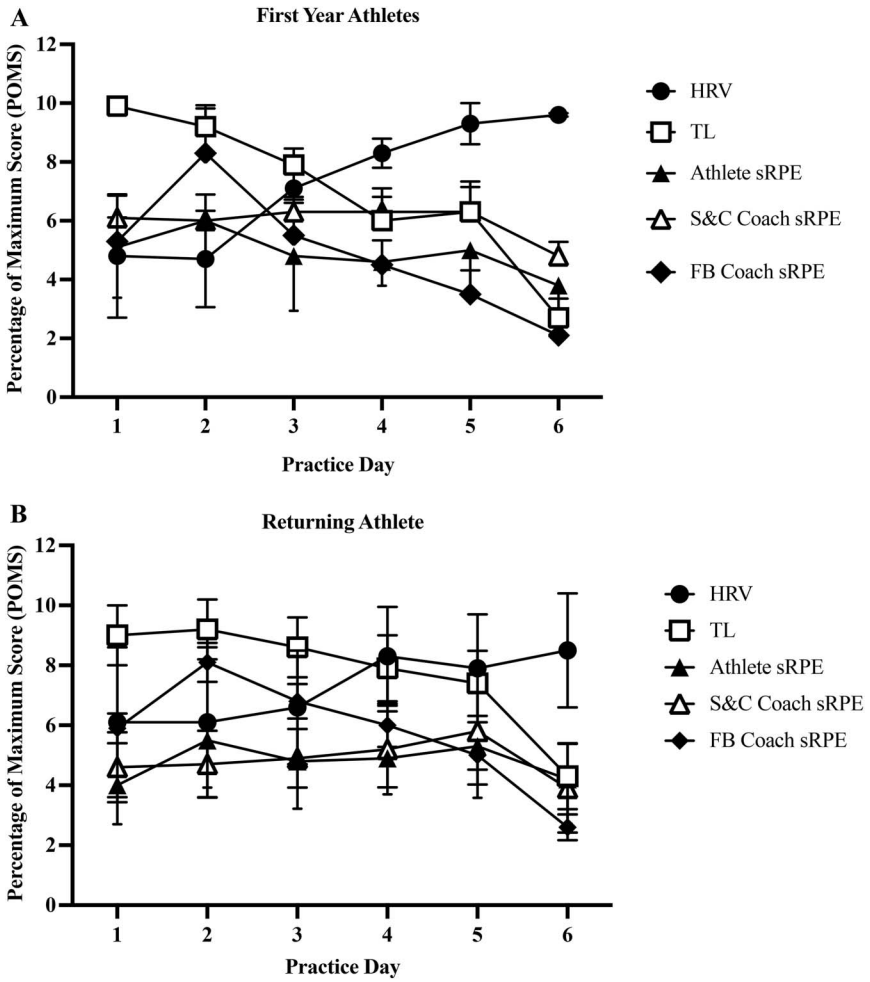

Figure 6. Visual comparison of first-year and returning players load and recovey.

training. This presents a second possible explanation, whereas coaches and athletes can both expect and more readily detect higher workout intensities in lesser fit athletes through the more obvious signs of mental and physical signs of distress and watch them more closely.

Finally, we explored the relationship between internal load and recovery and found that internal load was inversely related to our chosen HR-based recovery metrics (HRV). In a previous study, examining football athletes' external load and recovery, researchers found a more robust inverse relationship ($r = -0.60$) compared with this study's results (r = - 0.29) (14). It is well-documented that to improve performance and decrease the risk of either under recovery or overtraining, there must be an adequate balance of load followed by recovery (27). Recovery includes both physiological and psychological restorative processes (27), with the use of HRV becoming an increasingly popular metric due to its quick and noninvasive assessment (12).

There are several limitations of this study. First, the maximum HR values that were used to calculate TL were largely estimations (220-age) and not directly measured for all athletes (i.e., only when HR_{max} exceeded the calculated value, HR_{max} was "directly" measured). These estimations for HRmax may have artificially lowered our TRIMP scores (because the actual HR_{max} values were lower than the calculated values), adding error to both TL and POMS assessments. Second, the small and unequal sample sizes limited the strength of the comparative analyses (i.e., between first-year and returning athletes) and conclusions due to the potential for type II errors. However, because this was a preliminary investigation (with a limited budget), the investigators were unable to select a balanced number of subjects (i.e., the number of first-year and returning athletes were selected by the coaches for clinical reasons). In addition, only 2 coaches agreed to participate in this trial, which limited our ability to formulate robust conclusions. Nonetheless, we believe these pilot data are intriguing enough to launch further investigations on the topic of internal load quantification and subsequent load management. An ideal design would involve larger cohorts of American football athletes and their respective coaches, as the typical total number of coaches on a football team is 10 with 4 graduate assistants (24).

In conclusion, these exploratory data suggest that athletes and their coaches had a moderate degree of accuracy in perceiving daily internal load for each workout during the first week of preseason football training camp (as quantified by heart rate monitors). The FBC was better at observing the internal load for each athlete than the SCC. These relationships were stronger for the first-year athletes compared with the returning athletes to training camp. As training TL decreased across the first week of camp, recovery increased over time, as measured by the (patent protected) HR monitoring algorithms.

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Practical Applications

The overarching purpose of this study was to determine whether heart-rate monitoring systems enhanced our ability to assess individual athlete effort and recovery; particularly after long grueling football practices conducted during the peak summer heat. It seems that both objective (HR) and subjective (sRPE) measures of effort and recovery likely represent independent—but interdependent—modulators of stress. With practice and feedback about athletes' heart rates, both athletes and coaches may improve their TL perceptions and observations, respectively; such that their TL reports would align more closely with objectively measured TL. Athletes and coaches should continue to assess both objective and subjective measures of internal load, as both likely contribute unique insights into multiple stressors, which affect athlete load plus subsequent management.

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